

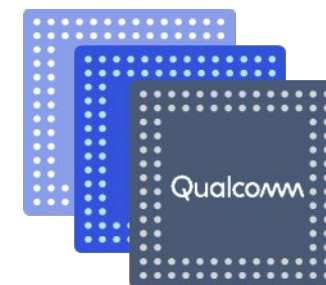
SM8650/SM8650P AI/ML Software Overview

80-40939-65 Rev. AB

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Revision History

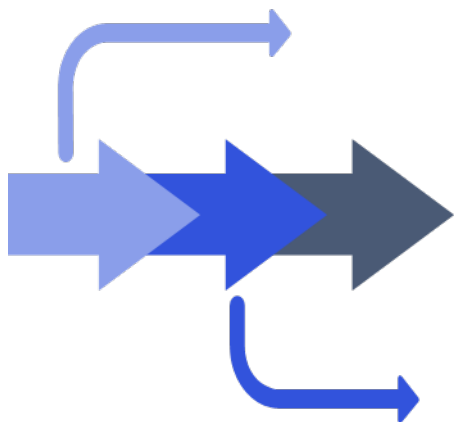
Revision	Date	Description
AA	January 2023	Initial release
AB	June 2023	Updated document title

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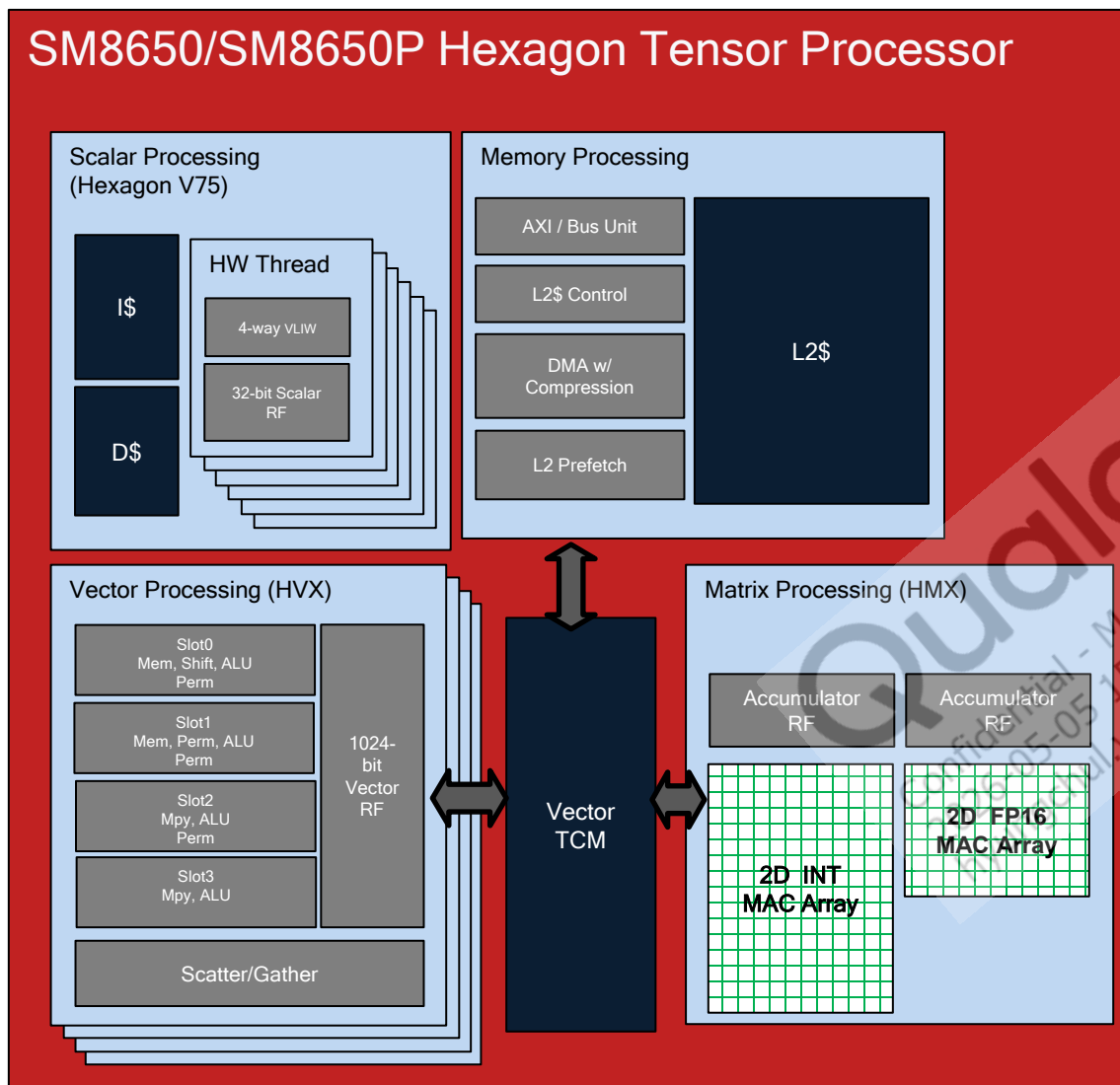
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Introduction to SM8650/SM8650P AI/ML Hardware

Qualcomm® Hexagon™ Tensor Processor on SM8650/SM8650P

SM8650/SM8650P Hexagon Tensor Processor

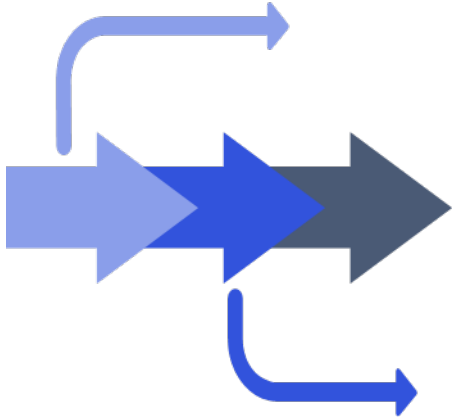


- The main design goal is improved power efficiency
- Features
 - V75 AI-optimized tensor processor
 - 6T scalar with enhanced micro-arch at higher $F_{\max}$
 - 4x HVX at higher $F_{\max}$
 - INT8/A16W8/A8W4 HMX + FP16 HMX
 - Large vector VTCM

See *Qualcomm Hexagon Tensor Processor Overview for SM8650* (80-40939-86) for more information

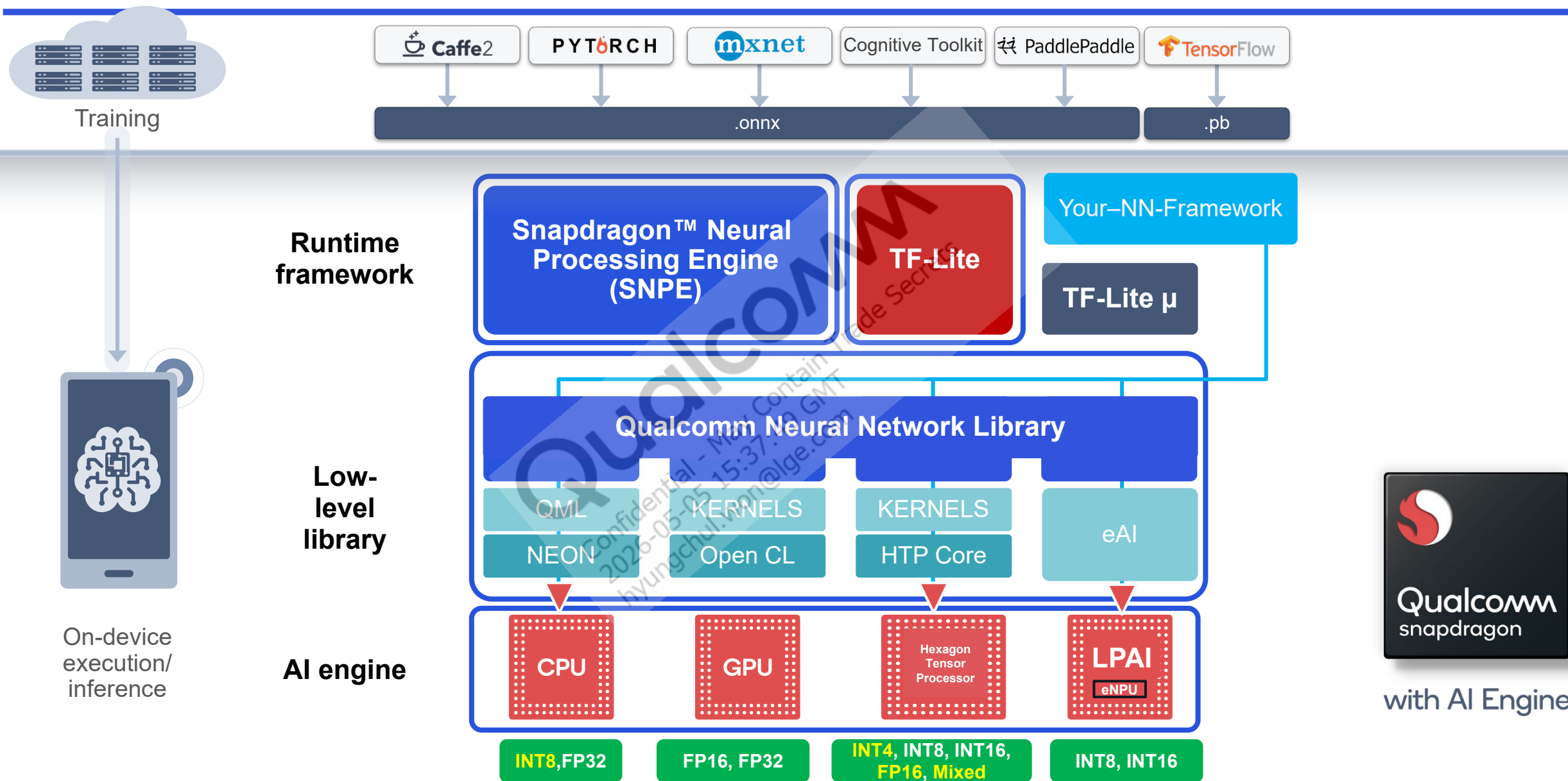
Low Power AI (Embedded NPU) on SM8650/SM8650P

	8550 – eNPU3 x2	8650 – Dual eNPU4 Architecture
Key Features	Model compression LSTM/GRU CNN/DW-CNN Fully connected	eNPU-3 Improved sparsity Native 4x8 Depthwise conv improvements Higher FMAX eNPU-4 (512 MAC) eNPU-3 Improved sparsity Native 4x8 Depthwise conv improvements Higher FMAX eNPU-4 (256 MAC)
Use Cases	Always On Camera Face Detection(OEM) , QR detection	Always On Camera Face Detection(OEM) , QR detection, Hand Gesture, Gaze detection
	Audio/Voice ML-based EC & NS, Embedded ASR/NLP/TTS, Q2Q Noise Suppression	Audio/Voice ML-based EC & NS, Embedded ASR/NLP/TTS, Q2Q Noise Suppression
	Qualcomm Sensing Hub Fall Detection In Car & Car Crash Device grip Detection Environment Detection	Qualcomm Sensing Hub Fall Detection In Car & Car Crash Device grip Detection Environment Detection
	Sensors Activity Recognition On/Off Body Backtap Double/Triple Tap	Sensors Activity Recognition On/Off Body Backtap Double/Triple Tap
Precision	16-bit Int, 8-bit Int	16-bit Int, 8-bit Int, 4-bit (weights)
Runtimes	eAI	eAI, eQNN



SM8550 AI/ML Software Solutions

SM8650/SM8650P AI Software Solution Overview



SM8650/SM8650P Changes for Hexagon Tensor Processor Software

Change	Description
Product unification (SNPE+QNN)	From SM8650/SM8650P, SNPE and Qualcomm Neural Network (QNN) SDKs will be released as a single, unified package
Distribution channel change	▪ From SM8650/SM8650P, AI software products will be released from the Qualcomm Package Manager ▪ CreatePoint releases will be deprecated
Hexagon Tensor Processor-specific improvements	▪ Model preparation improvements ▪ Improved pre-emption for concurrent use cases ▪ VTCM partitioning and sharing ▪ Thread reservation ▪ Additional data type mixing in a single model

SM8650/SM8650P AI Software SDKs

- Design objective: Abstract developers from hardware complexities
- Features:
 - Combined package for SNPE SDK and QNN SDK
 - Support for SM8650/SM8650P and other devices in multiple tiers
 - Hardware IP and supported precisions on SM8650/SM8650P:
 - CPU: INT8, FP32
 - GPU: FP16, FP32
 - HTP: INT4, INT8, INT16 (16a8w), INT16 (16a16w)*, FP16
 - Mixed: INT4-INT8-INT16-FP16
- Enablement:
 - EAR for OEMs/ISVs one month ahead of SM8650/SM8650P FC
 - General availability by 2023 Q4

** For selected ops only*

Qualcomm Neural Network Delegate SDK

- Design objective: Enable the Android ecosystem for TensorFlow Lite models
- Google NNAPI delegate: Removed per guidance from Google
- QNN delegate SDK: Support for SM8650/SM8650P and other devices in multiple tiers
- Enablement:
 - EAR for OEMs/ISVs to intercept SM8650/SM8650P CS
 - General availability by 2023 Q4

Qualcomm Tensor Virtual Machine SDK

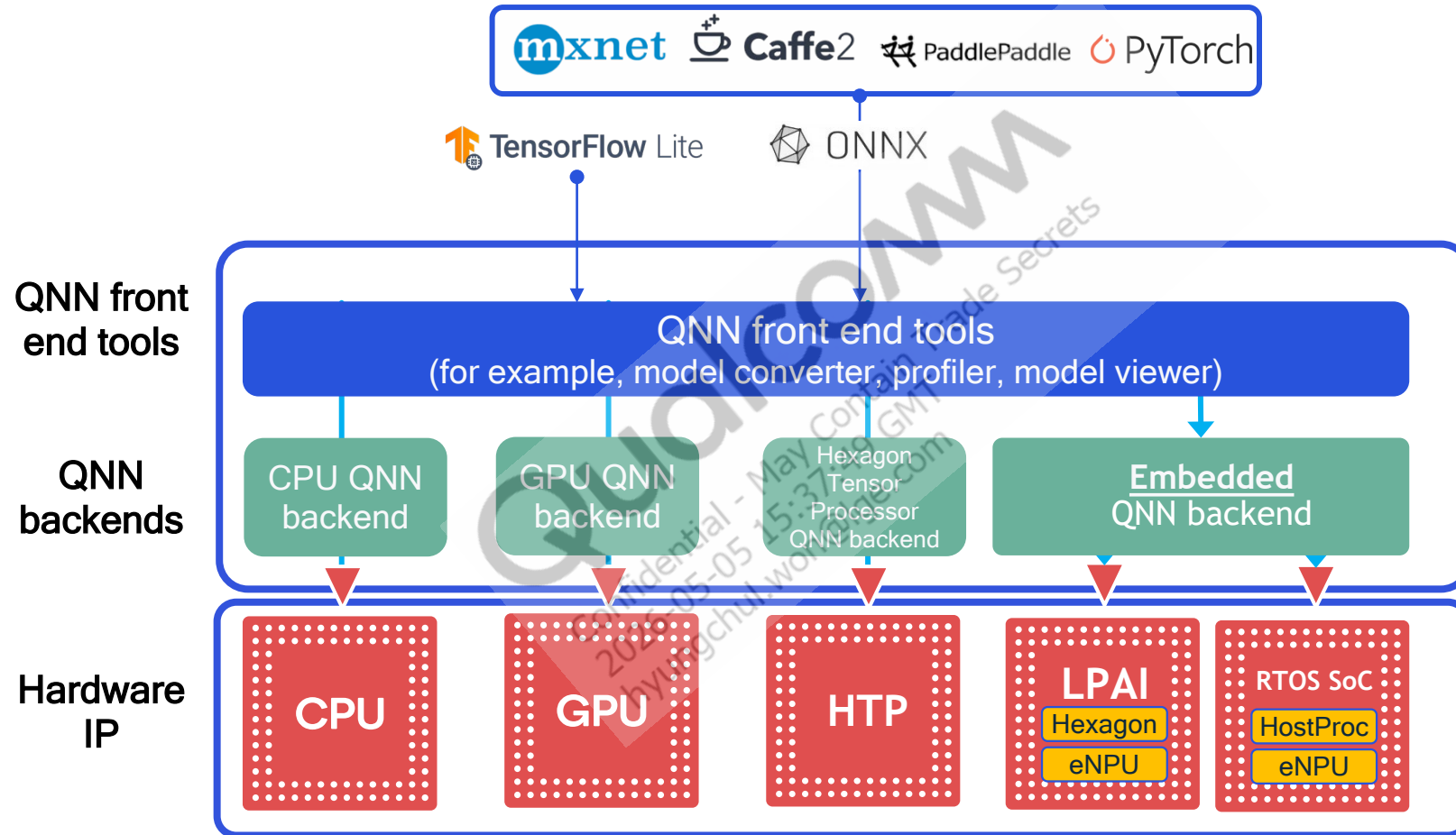
- Design objective: Improve developer productivity
- Features:
 - Support for SM8650/SM8650P and other devices in multiple tiers
 - Open-source, end-to-end compilation stack for deep learning
 - Support for custom ops generation for SNPE and QNN for the INT8, INT16 (16a8w), INT16 (16a16w) data types
 - Will support scalar only or scalar + HVX code generation to support Hexagon architectures
 - Will support generation of ops for eNPU
- Enablement:
 - EAR for OEMs/ISVs to intercept SM8650/SM8650P FC
 - General availability 2023 Q4

AI Tools

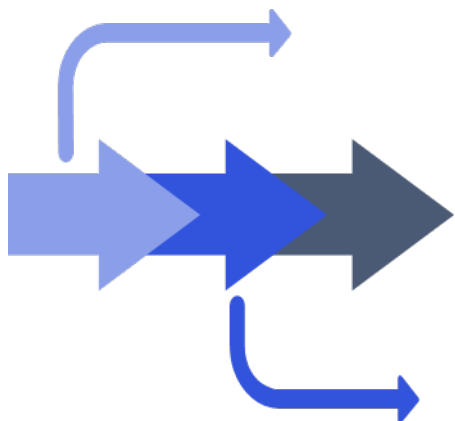
- Design objective: Improve developer productivity
- AI model efficiency toolkit:
 - Include eNPU support, add ONNX PTQ
 - Tighter integration with SNPE/QNN Inferencing accuracy
 - Supporting additional mixing of precision for QAT
- Unifying AI tools for customer productivity:
 - Architecture checker
 - Quantization checker, evaluator, debugger
 - Performance analyzer
 - Hexagon Tensor Processor simulator (bit accurate)
 - Adding HVX, HMX, VTCM, and DDR use
- Neural architecture search (NAS): Better integration between QNN and Google Vertex NAS

Embedded AI ML – Tools Convergence

QNN is extended to the embedded AI runtime*



* eAI SDK will still be supported while extension to QNN is made



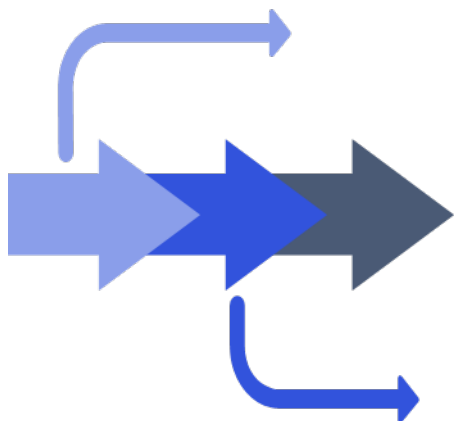
References

References

Documents	
Title	Number or link
Qualcomm Technologies, Inc.	
Qualcomm Hexagon Tensor Processor Overview for SM8650	80-40939-86
Qualcomm AI Model Efficiency Toolkit Pro Feature Overview	80-PK882-19
Qualcomm Snapdragon Neural Processing Engine (SNPE) Overview	80-NL315-18
Resources	
Qualcomm CreatePoint	https://createpoint.qti.qualcomm.com/
Apache TVM Stack	https://tvm.apache.org/

References (cont.)

Acronyms	
Acronym or term	Definition
eAI	Embedded AI
eNPU	Embedded Neural Processing Unit
HVX	Hexagon Vector eXtensions
LPAI	Low power AI
NAS	Neural architecture search
NN	Neural network
QNN	Qualcomm Neural Network
SNPE	Snapdragon Neural Processing Engine
TVM	Tensor Virtual Machine
UDO	User-defined object
VTM	Vector TCM



Questions?

<https://createpoint.qti.qualcomm.com>
